*Proof of (H1).* One can readily see that $G^{(n+1)} = \{e\}$ since $\pi_1(W_0)^{(n+1)} \subset \mathcal{P}^{n+1}\pi_1(W_0)$. For $\Gamma = \pi_1(W_0)$ and $0 \le k \le n$, $\mathcal{P}^k\Gamma/\mathcal{P}^{k+1}\Gamma$ injects into $H_1\left(\Gamma; R\left[\Gamma/\mathcal{P}^k\Gamma\right]\right)$ with $R = \mathbb{Q}$ or $\mathbb{Z}_{p_1}$, which is an $R$-vector space. So, by Lemma 3.4(2), $G$ lies in Strebel's class $D(\mathbb{Z}_{p_1})$ (see also [Cha13, Lemma 4.3]).

Also, a meridian of $J$ generates

$$\pi_1(W_0)/\mathcal{P}^1\pi_1(W_0) \cong H_1(W_0)/\text{torsion} = H_1(W_0) \cong H_1(M(J)) = \mathbb{Z}$$

onto which $G$ surjects. So $\phi$ sends a meridian of $J$ to an infinite order element in $G$.

*Proof of (H2).* Let $\mu_k^{i,j}$ be a meridian of $M(J_k^i)^j$. Since $H_1(M(J_0^i)^j) \cong \mathbb{Z}$ is generated by $\mu_0^{i,j}$, and since the group $\mathcal{P}^n\pi_1(W_0)/\mathcal{P}^{n+1}\pi_1(W_0)$ is a $\mathbb{Z}_{p_1}$-vector space, $\phi$ satisfies (H2) if it satisfies (H2′) below.

(H2′) The image of each $\mu_0^{i,j}$ under $\phi$ is nontrivial in $\mathcal{P}^n\pi_1(W_0)/\mathcal{P}^{n+1}\pi_1(W_0)$.

For brevity, we write $\mu_k$ for $\mu_k^{i,j}$ when the superscripts are understood. We will show, inductively, that for $k = n, n-1, \dots, 0$, $\mu_k = \mu_k^{i,j}$ is nontrivial in $\mathcal{P}^{n-k}\pi_1(W_k)/\mathcal{P}^{n-k+1}\pi_1(W_k)$. The $k = 0$ case implies the desired conclusion (H2′).

For $k = n$, the claim follows from

$$\mathcal{P}^0\pi_1(W_n)/\mathcal{P}^1\pi_1(W_n) \cong H_1(W_n) \cong H_1(M(J_n^i)^j) \cong \langle \mu_n \rangle \cong \mathbb{Z}.$$

Suppose that $\mu_{k+1} = \mu_{k+1}^{i,j}$ is nontrivial in $\mathcal{P}^{n-k-1}\pi_1(W_{k+1})/\mathcal{P}^{n-k}\pi_1(W_{k+1})$, $k \le n-1$. We will show that $\mu_k$ is nontrivial in the quotient $\mathcal{P}^{n-k}\pi_1(W_k)/\mathcal{P}^{n-k+1}\pi_1(W_k)$.

First, we show that $\mu_k \in \mathcal{P}^{n-k}\pi_1(W_k)$. Recall that $M_k^{i,j} \subset \partial E_k^{i,j}$ is a copy of $M(P(U))$. Let $\eta_k = \eta_k^{i,j} \subset M_k^{i,j}$ be the corresponding copy of the axis of $P$. Since $\eta_k$ is identified with $\mu_k$ in $M(J_{k+1}^i)^j$ and $P$ has winding number zero, we have $\mu_k \in \langle \mu_{k+1} \rangle^{(1)}$, where $\langle - \rangle$ denotes the subgroup of $\pi_1(W_k)$ normally generated by $-$. Inductively, it follows that $\mu_k \in \langle \mu_n \rangle^{(n-k)}$. Therefore, $\mu_k \in \pi_1(W_k)^{(n-k)} \subset \mathcal{P}^{n-k}\pi_1(W_k)$. (See also the last paragraph of [Cha13, p. 4799].)

Next, we assert that

$$\mathcal{P}^{n-k}\pi_1(W_k)/\mathcal{P}^{n-k+1}\pi_1(W_k) \cong \mathcal{P}^{n-k}\pi_1(W_{k+1})/\mathcal{P}^{n-k+1}\pi_1(W_{k+1}). \tag{4.4}$$

By arguments in [Cha13, Assertion 1 on p. 4799], we have

$$\pi_1(W_k)/\pi_1(W_k)^{(n-k+1)} \cong \pi_1(W_{k+1})/\pi_1(W_{k+1})^{(n-k+1)}. \tag{4.5}$$

Also, from the definitions of $\mathcal{P}^i\Gamma$, one can see that if a homomorphism $A \to B$ induces $A/A^{(j)} \cong B/B^{(j)}$, then the homomorphism also induces $\mathcal{P}^i A \cong \mathcal{P}^i B$ for $i \le j$. Therefore, the inclusion $W_{k+1} \hookrightarrow W_k$ induces $\mathcal{P}^i\pi_1(W_k) \cong \mathcal{P}^i\pi_1(W_{k+1})$ for $i \le n-k+1$, and (4.4) follows.

By (4.4), we only need to show that $\mu_k$ is nontrivial in $\mathcal{P}^{n-k}\pi_1(W_{k+1})/\mathcal{P}^{n-k+1}\pi_1(W_{k+1})$. Let $\pi = \pi_1(W_{k+1})$ and $\Gamma = \pi/\mathcal{P}^{n-k}\pi$. Let $R = \mathbb{Q}$ when $k > 0$ and $R = \mathbb{Z}_{p_1}$ when $k = 0$. Since $\mathcal{P}^{n-k}\pi/\mathcal{P}^{n-k+1}\pi$ injects into $H_1(W_{k+1}; R[\Gamma])$, it suffices to show that $\mu_k$ is nontrivial in $H_1(W_{k+1}; R[\Gamma])$.

To show this, we use the (higher order) Blanchfield forms, following the ideas in [CHL09] and [Cha13]. Denote the Blanchfield form of $M(J_{k+1}^i)^j$ over $R[t^{\pm 1}]$ by

$$B\ell^R \colon H_1(M(J_{k+1}^i)^j; R[t^{\pm 1}]) \times H_1(M(J_{k+1}^i)^j; R[t^{\pm 1}]) \longrightarrow R(t)/R[t^{\pm 1}].$$

Also, let

$$\mathcal{B} \colon H_1(M(J_{k+1}^i)^j; R[\Gamma]) \times H_1(M(J_{k+1}^i)^j; R[\Gamma]) \longrightarrow \mathcal{K}/R[\Gamma]$$

be the higher order Blanchfield form of $M(J_{k+1}^i)^j$ over $R[\Gamma]$ defined in [COT03, Theorem 2.13]. Here, $\mathcal{K}$ is the (skew) field of quotients of $R[\Gamma]$: since $\mathcal{P}^i\pi/\mathcal{P}^{i+1}\pi$ is torsion-free for $0 \le i \le n-1$ by the definition of $\mathcal{P}^{i+1}\pi$ and since the range of $k$ is $0, \dots, n-1$, the group $\Gamma = \pi/\mathcal{P}^{n-k}\pi$ is poly-torsion-free-abelian. (If $k$ were $-1$, $\pi/\mathcal{P}^{n+1}\pi$ would be allowed to have $p$-torsion.) It

follows that $R[\Gamma]$ embeds into its (skew) field of quotients $\mathcal{K}$, due to [COT03, Proposition 2.5] when $R = \mathbb{Q}$ (i.e. $k > 0$) and [Cha13, Lemma 5.2] when $R = \mathbb{Z}_{p_1}$ (i.e. $k = 0$).

We assert that $\mathcal{B}(\mu_k, \mu_k) \neq 0$. By our induction hypothesis that the meridian $\mu_{k+1}$ is nontrivial in the torsion-free abelian group $\mathcal{P}^{n-k-1}\pi/\mathcal{P}^{n-k}\pi \subset \Gamma$, one can see that

$$H_1(M(J^i_{k+1})^j; R[\Gamma]) \cong R[\Gamma] \otimes_{R[t^{\pm 1}]} H_1(M(J^i_{k+1})^j; R[t^{\pm 1}]). \tag{4.6}$$

By [Lei06, Theorem 4.7] and [Cha13, Theorem 5.4], we have $\mathcal{B}(1 \otimes x, 1 \otimes x) = 0$ if and only if $B\ell^R(x, x) = 0$. Also by our assumption on $\eta_k$ and choice of $p_1$, we have $B\ell^R(\eta_k, \eta_k) \neq 0$. Since $\eta_k$ is identified with $\mu_k$ in $M(J^i_{k+1})^j$, it follows that $B\ell^R(\mu_k, \mu_k) \neq 0$, and hence $\mathcal{B}(1 \otimes \mu_k, 1 \otimes \mu_k) \neq 0$. Therefore, viewing $\mu_k = 1 \otimes \mu_k$ as an element in $H_1(M(J^i_{k+1})^j; R[\Gamma])$ via the isomorphism (4.6), it follows that $\mathcal{B}(\mu_k, \mu_k) \neq 0$.

Now we invoke the following lemma, which will be proved in Section 4.3.

**Lemma 4.8.** *The pair $(W_{k+1}, \phi\colon \pi_1(W_{k+1}) \to \Gamma)$ is an $R$-coefficient Blanchfield bordism.*

Here, we use the notion of an *$R$-coefficient Blanchfield bordism* in the sense of [Cha14, Definition 4.11]. The only property of a Blanchfield bordism we need to complete the proof of Proposition 4.6 is that for the kernel

$$M = \mathrm{Ker}\{H_1(M(J^i_{k+1})^j; R[\Gamma]) \to H_1(W_{k+1}; R[\Gamma])\}$$

of the inclusion-induced map, we have $\mathcal{B}(x, x) = 0$ for all $x \in M$ (see [Cha14, Theorem 4.12]).

From this, it follows that $\mu_k \notin M$ since $\mathcal{B}(\mu_k, \mu_k) \neq 0$. Therefore, $\mu_k$ is nontrivial in $H_1(W_{k+1}; R[\Gamma])$. This completes the proof of Proposition 4.6 modulo the proof of Lemma 4.8, which is given below.

## 4.3. Lagrangian bordism and Blanchfield bordism

In this subsection, we give a proof of Lemma 4.8. We also discuss a generalization of Theorem A at the end of the subsection.

Throughout this subsection, we suppose the following: let $R = \mathbb{Z}_p$ or $\mathbb{Q}$. Let $W$ be a compact connected 4-manifold with boundary and $\psi\colon \pi_1(W) \to G$ be a homomorphism where $G$ is a poly-torsion-free-abelian group. Let $\mathcal{K}G$ be the (skew) field of quotients of $RG$. Let $\lambda_G$ be the equivariant intersection form

$$\lambda_G\colon H_2(W; RG) \times H_2(W; RG) \longrightarrow RG.$$

The following definition is inspired by the notion of an $n$-bordism in the sense of [CHL09, Definition 5.2] and [Cha13, Definition 5.5] and by [Cha14, Theorem 4.13].

**Definition 4.9.** A pair $(W, \psi)$ is an *$R$-coefficient Lagrangian bordism* if it satisfies the following.

1. There exist elements $\ell_1, \dots, \ell_m,\ d_1, \dots, d_m \in H_2(W; RG)$ such that $\lambda_G(\ell_i, \ell_j) = 0$ and $\lambda_G(\ell_i, d_j) = \delta_{ij}$, $1 \le i, j \le m$.
2. $\mathrm{rank}_R\, \mathrm{Coker}\{H_2(\partial W; R) \to H_2(W; R)\} \le 2m$.

If $W$ is an integral $n$-solution, then $H_2(\partial W; R) \to H_2(W; R)$ is trivial. Therefore, the lemma below immediately follows from the definition of an integral $n$-solution (see [Cha13, Definition 3.1]).

**Lemma 4.10.** *Suppose that $W$ is an integral $n$-solution. Then, for any homomorphism $\psi\colon \pi_1(W) \to G$ that factors through $\pi_1(W)/\pi_1(W)^{(n)}$, the pair $(W, \psi)$ is an $R$-coefficient Lagrangian bordism.*

Recall that Lemma 4.8 asserts that $(W_{k+1}, \phi)$ is an $R$-coefficient Blanchfield bordism. In the proof of Lemma 4.8, we do not directly verify that $(W_{k+1}, \phi)$ satisfies the definition of an $R$-coefficient Blanchfield bordism given in [Cha14, Definition 4.11] but use the following fact.

**Proposition 4.11.** *Suppose that* $\operatorname{rank}_R H_1(M;R) = 1$ *for each component* $M$ *of* $\partial W$*. If* $(W,\psi)$ *is an* $R$*-coefficient Lagrangian bordism, then it is an* $R$*-coefficient Blanchfield bordism.*

*Proof.* By [Cha14, Lemma 4.15], $(W,\psi)$ is an $R$-coefficient Blanchfield bordism if the following hold.

(1) There exist elements $\ell_1,\dots,\ell_m,\ d_1,\dots,d_m \in H_2(W;RG)$ such that $\lambda_G(\ell_i,\ell_j) = 0$ and $\lambda_G(\ell_i,d_j) = \delta_{ij}$, $1 \le i,j \le m$.
(2) $\dim_{\mathcal{K}G} \operatorname{Coker}\{H_2(\partial W;\mathcal{K}G) \to H_2(W;\mathcal{K}G)\} \le 2m$.

Definition 4.9 (1) implies (1). Definition 4.9 (2) implies (2) by Lemma 4.12 below. □

**Lemma 4.12.** *Suppose* $\operatorname{rank}_R H_1(M;R) = 1$ *for each component* $M$ *of* $\partial W$*. Then,*

$$\dim_{\mathcal{K}G} \operatorname{Coker}\{H_2(\partial W;\mathcal{K}G) \to H_2(W;\mathcal{K}G)\} \le \operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W;R) \to H_2(W;R)\}.$$

*Proof.* In [CHL09, Lemma 5.10 (1)], it was shown that the equality

$$\dim_{\mathcal{K}G} \operatorname{Coker}\{H_2(\partial W;\mathcal{K}G) \to H_2(W;\mathcal{K}G)\} = \operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W;R) \to H_2(W;R)\}.$$

holds under a stronger hypothesis. Examining the proof of [CHL09, Lemma 5.10 (1)], which analyzes the long exact sequences of $(W,\partial W)$ with coefficients in $R$ and $\mathcal{K}G$, one sees that the same argument shows the desired inequality under our weaker hypothesis. For the reader's convenience, we describe details below. We remark that we do not use that $W$ is an $R$-coefficient Lagrangian bordism in this proof.

If $\psi\colon \pi_1(W) \to G$ is trivial, then one can readily obtain the desired inequality (actually equality in this case). Suppose that $\psi$ is nontrivial. For brevity, let

$$r = \operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W;R) \to H_2(W;R)\}$$
$$r' = \dim_{\mathcal{K}G} \operatorname{Coker}\{H_2(\partial W;\mathcal{K}G) \to H_2(W;\mathcal{K}G)\}$$

and denote by $\beta_i$ and $b_i$ be the $i$th Betti numbers with $R$-coefficients and $\mathcal{K}G$-coefficients, respectively. Using the homology long exact sequence for $(W,\partial W)$ with $R$-coefficients and the Poincaré duality $\beta_i(W) = \beta_{4-i}(W,\partial W)$, one can readily see that

$$r = \chi(W) + 2\beta_1(W) - 2 - \beta_1(\partial W) + \beta_0(\partial W),$$

where $\chi$ is the Euler characteristic. For the $\mathcal{K}G$-coefficient case, since $\psi$ is nontrivial, $b_0(W) = 0$ by [COT03, Propositions 2.9], and the the same argument gives

$$r' = \chi(W) + 2b_1(W) - b_1(\partial W) + b_0(\partial W).$$

Therefore,

$$r - r' = 2(\beta_1(W) - 1 - b_1(W)) - (\beta_1(\partial W) - b_1(\partial W)) + (\beta_0(\partial W) - b_0(\partial W)).$$

Since $\psi$ is nontrivial, we have $\beta_1(W) - 1 - b_1(W) \ge 0$ by [COT03, Proposition 2.11]. So, to complete the proof, it suffices to show that for each component $M$ of $\partial W$

$$\beta_1(M) - b_1(M) = \beta_0(M) - b_0(M).$$

It holds if $\psi$ induces a trivial homomorphism of $\pi_1(M)$, since $\beta_i(M) = b_i(M)$ in this case. Otherwise, our hypothesis $\beta_1(M) = 1$ implies that $b_1(M) = 0$ by [COT03, Propositions 2.11]. Also $b_0(M) = 0$ by [COT03, Propositions 2.9]. Since $\beta_0(M) = 1$, the promised equality holds. □

In the following lemma, the main examples to keep in mind are $W = V$ and $X = C$ or $E^i_k$ in the proof of Theorem A.

**Lemma 4.13.** *Let* $W$ *and* $X$ *be compact connected 4-manifolds with boundary. Suppose* $\partial X = \partial_1 X \sqcup \partial_2 X$ *where each* $\partial_i X$ *is a union of components of* $\partial X$ *and* $W \cap X = \partial_1 X$*. Let* $W' = W \cup X$ *and* $\psi\colon \pi_1(W') \to G$ *be a homomorphism. Suppose the following:*

(1) *the inclusion induces an injection* $H_i(\partial_1 X; R) \to H_i(X; R)$ *for* $i = 1, 2$, *and*
(2) *induces a surjection* $H_2(\partial_2 X; R) \to H_2(X; R)$.

*If additionally* $(W, \psi|_{\pi_1(W)})$ *is an* $R$*-coefficient Lagrangian bordism,* $(W', \psi)$ *is an* $R$*-coefficient Lagrangian bordism.*

*Proof.* If $\ell_1, \dots, \ell_m$, $d_1, \dots, d_m$ are elements in $H_2(W; RG)$ such that $\lambda_G(\ell_i, \ell_j) = 0$ and $\lambda_G(\ell_i, d_j) = \delta_{ij}$, then they satisfy the same intersection form conditions on $H_2(W'; RG)$.

By (1) and (2) and an application of the Mayer–Vietoris sequence associated with $W' = W \cup X$, one can readily see that

$$\operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W; R) \to H_2(W; R)\} \geq \operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W'; R) \to H_2(W'; R)\}.$$

Therefore, $\operatorname{rank}_R \operatorname{Coker}\{H_2(\partial W'; R) \to H_2(W'; R)\} \leq 2m$. □

Recall that for knots $K_1, \dots, K_n$, there is a standard cobordism $C = C(a_1K_1, \dots, a_nK_n)$ such that

$$\partial_- C = M(a_1K_1 \# \cdots \# a_nK_n) \quad \text{and} \quad \partial_+ C = \bigsqcup_{i=1}^{n} a_i M(K_i).$$

Also recall that for a knot $K$ and a satellite operator $P$ with axis $\eta$, there is a cobordism $E(P, K)$ such that

$$\partial_- E(P, K) = M(P(K)) \quad \text{and} \quad \partial_+ E(P, K) = M(K) \sqcup M(P(U)).$$

The following lemma is well known (e.g. see the first paragraph of [COT04, p. 114] and [CHL09, Lemma 2.5(3)]), and it follows from a direct computation of the relevant homology cobordism groups using relative chain complexes.

**Lemma 4.14.** *Let* $X = C$ *or* $E(P, K)$ *as above. Then, the inclusions induce an injection* $H_i(\partial_- X; R) \to H_i(X; R)$ *for* $i = 1, 2$, *and a surjection* $H_2(\partial_+ X; R) \to H_2(X; R)$.

Now we are ready to prove Lemma 4.8.

*Proof of Lemma 4.8.* Let $\pi = \pi_1(W_{k+1})$ and $\Gamma = \pi/\mathcal{P}^{n-k}\pi$. Recall that Lemma 4.8 states that the pair $(W_{k+1}, \psi\colon \pi \to \Gamma)$ is an $R$-coefficient Blanchfield bordism.

We have

$$W_{k+1} = V \cup C \cup \left(\bigsqcup_{i=1}^{m'} a_i E^i_{n-1}\right) \cup \left(\bigsqcup_{i=1}^{m'} a_i E^i_{n-2}\right) \cup \cdots \cup \left(\bigsqcup_{i=1}^{m'} a_i E^i_{k+1}\right).$$

Since $V$ is an integral $(n+1)$-solution, it is also an integral $(n-k)$-solution. We have $\Gamma^{n-k} = \{e\}$ since $\pi^{n-k} \subset \mathcal{P}^{n-k}\pi$. Therefore, it follows from Lemma 4.10 that $(V, \psi)$ is an $R$-coefficient Lagrangian bordism. One can readily see that $(W_{k+1}, \psi)$ is an $R$-coefficient Lagrangian bordism by repeatedly applying Lemmas 4.13 and 4.14 with $X = C$ or $X = E^i_\ell$, $1 \leq i \leq m'$ and $k+1 \leq \ell \leq n-1$.

Now, $(W_{k+1}, \psi)$ is an $R$-coefficient Blanchfield bordism by Proposition 4.11 since we have $\operatorname{rank}_R H_1(M; R) = 1$ for each boundary component $M$ of $W_{k+1}$. □

**Generalization of Theorem A.** Theorem A generalizes to compositions of distinct satellite operators. The proof of Theorem A does not use specific properties of the pattern $P(U)$, and this leads us to the following theorem.

**Theorem 4.15** (Generalization of Theorem A)**.** *Let* $n \geq 1$. *For* $1 \leq k \leq n$, *let* $P_k\colon \mathcal{C} \to \mathcal{C}$ *be a winding number zero satellite operator with axis* $\eta_k$. *Let* $B\ell_k$ *be the Blanchfield form of* $P_k(U)$. *If* $B\ell_k(\eta_k, \eta_k)$ *is nontrivial for* $1 \leq k \leq n$, *then for any winding number zero satellite operator* $P_{n+1}\colon \mathcal{C} \to \mathcal{C}$, *the quotient*

$$\langle (P_n \circ P_{n-1} \circ \cdots \circ P_1)(\mathcal{C}) \rangle / \langle (P_{n+1} \circ P_n \circ \cdots \circ P_1)(\mathcal{C}) \rangle$$